EXP 4

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clc;
clear;
close all;

% MATLAB Simulation of Superheterodyne Receiver for AM
% Communication System with Frequency Mixing and Filtering

%% 1. System Parameters
Fs = 500000;          % Sampling frequency (Hz)
t = 0:1/Fs:0.02;      % Time vector

fm = 1000;            % Message frequency (Hz)
fc = 100000;          % RF carrier frequency (Hz)
fLO = 110000;         % Local oscillator frequency (Hz)
fIF = abs(fLO - fc);  % Intermediate frequency (Hz)

Am = 1;               % Message amplitude
Ac = 1;               % Carrier amplitude
mu = 0.7;             % Modulation index

%% 2. Generate Message Signal
m = Am*cos(2*pi*fm*t);

%% 3. Generate AM Signal
sAM = Ac*(1 + mu*m/Am).*cos(2*pi*fc*t);

%% 4. Frequency Mixing
% Mixer produces sum and difference frequencies
localOscillator = cos(2*pi*fLO*t);
mixerOutput = 2*sAM.*localOscillator;
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%% 5. Intermediate Frequency (IF) Filtering
% IF = 10 kHz
% Select frequencies around the IF carrier

N = length(t);
f = (-floor(N/2):ceil(N/2)-1)*(Fs/N);

MIX_F = fftshift(fft(mixerOutput));

% Ideal bandpass filter centered at +/- IF
IF_BW = 6000;
HIF = double(abs(abs(f)-fIF) <= IF_BW/2);

IF_F = MIX_F.*HIF;
IF_signal = real(ifft(ifftshift(IF_F)));

%% 6. AM Demodulation Using Envelope Detection
% Hilbert transform gives the signal envelope
analyticSignal = hilbert(IF_signal);
envelope = abs(analyticSignal);

% Remove DC component
recovered = envelope - mean(envelope);

% Normalize recovered signal
recovered = recovered/max(abs(recovered));

%% 7. Plot Time-Domain Signals
figure('Name','Superheterodyne AM Receiver');
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subplot(5,1,1);
plot(t,m,'b');
title('Original Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(5,1,2);
plot(t,sAM,'r');
title('AM Modulated RF Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(5,1,3);
plot(t,mixerOutput,'k');
title('Mixer Output');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(5,1,4);
plot(t,IF_signal,'m');
title('Intermediate Frequency (IF) Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(5,1,5);
plot(t,recovered,'g','LineWidth',1.2);
title('Recovered Message Signal');
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xlabel('Time (s)');
ylabel('Amplitude');
grid on;

%% 8. Frequency Spectrum
figure('Name','Frequency Spectrum');

subplot(3,1,1);
plot(f/1000,abs(fftshift(fft(sAM)))/N);
xlim([0 150]);
title('AM Signal Spectrum');
xlabel('Frequency (kHz)');
ylabel('Magnitude');
grid on;

subplot(3,1,2);
plot(f/1000,abs(MIX_F)/N);
xlim([0 250]);
title('Mixer Output Spectrum');
xlabel('Frequency (kHz)');
ylabel('Magnitude');
grid on;

subplot(3,1,3);
plot(f/1000,abs(IF_F)/N);
xlim([0 30]);
title('IF Filter Output Spectrum');
xlabel('Frequency (kHz)');
ylabel('Magnitude');
grid on;
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%% 9. Display Parameters
fprintf('RF Carrier Frequency = %.2f kHz\n',fc/1000);
fprintf('Local Oscillator Frequency = %.2f kHz\n',fLO/1000);
fprintf('Intermediate Frequency = %.2f kHz\n',fIF/1000);
fprintf('Message Frequency = %.2f kHz\n',fm/1000);

disp('Superheterodyne AM receiver simulation completed.');
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